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MIL-M-51500A(EA)
21 June 1988
SUPERSEDING
MIL-M-51500(EA)
17 April 1985

MILITARY SPECIFICATION

METHYLPHOSPHONIC DIFLUORIDE, TECHNICAL

This specification is approved for use within U.S. Army Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of technical grade methylphosphonic difluoride (also known as difluoromethylphosphine oxide).

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPT-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC NO. A4436 FSC 6810

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1.2 Classification. Methylphosphonic difluoride shall be of the following types as specified (see 6.2):

Type I - Unstabilized
Type II - Stabilized

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

BB-N-411 - Nitrogen, Technical
NN-P-71 - Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
PPP-C-2020 - Chemicals, Liquid, Dry, and Paste: Packaging of
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-T-60 - Tape: Packaging, Waterproof

MILITARY

MIL-B-26701 - Bottles, Screw Cap and Carboys; Polyethylene Plastic
MIL-D-51161 - N,N'-Diisopropylcarbodiimide

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-1168 - Ammunition Lot Numbering

2.1.2 Other Government documents and drawings. The following other Government documents and drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171 to 199 - Department of Transportation Hazardous Materials Regulations

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(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. Orders for the above publication should cite "49 CFR 171 to 199.")

DRAWINGS

U.S. ARMY CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

400-71-8 - Methylphosphonic Difluoride, Packing of

(Copies of specifications, standards, other Government documents, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ASTM STANDARDS

C 516 - Vermiculite Loose Fill Thermal Insulation
D 1193 - Reagent Water

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Type I methylphosphonic difluoride.

3.1.1 Purity by freezing point depression analysis. Type I methylphosphonic difluoride shall have a purity of no less than 98.0 mole percent when tested as specified in 4.4.4.1.

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3.1.2 Proton nuclear magnetic resonance (NMR) response. When tested by proton NMR analysis as specified in 4.4.4.2, type I methylphosphonic difluoride shall conform to the purity and identification characteristics specified below. If paramagnetic impurities interfere with the proton NMR analysis, the type I methylphosphonic difluoride shall be identified by its infrared spectrum characteristics as specified in 3.1.3.

3.1.2.1 Purity. Type I methylphosphonic difluoride shall have a purity as determined by proton NMR analysis of no less than 98.0 mole percent.

3.1.2.2 Identification characteristics. A proton NMR spectrum of type I methylphosphonic difluoride shall consist of a doublet of triplets. The coupling constant (hydrogen-phosphorus) shall be 19.8 $\pm$ 0.3 Hertz (Hz) and the coupling constant (hydrogen-fluorine) shall be 6.2 $\pm$ 0.3 Hz.

3.1.3 Infrared spectrum characteristics. (This requirement shall be applicable only when paramagnetic impurities interfere with the proton NMR analysis specified in 3.1.2). An infrared spectrum of type I methylphosphonic difluoride shall essentially consist of all the infrared absorption peaks, at the same frequencies and in the same relative intensities, as a standard infrared spectrum of unstabilized methylphosphonic difluoride (figure 1) when tested as specified in 4.4.4.3. An infrared spectrum with a single broad band at 1322 centimeters^{-1} (cm^{-1}) instead of two bands at 1335 cm^{-1} and 1310 cm^{-1} shall be acceptable provided that all other infrared spectrum characteristics conform to specified requirements.

3.2 Type II methylphosphonic difluoride. Type II methylphosphonic difluoride shall consist of type I methylphosphonic difluoride conforming to 3.1 to which 2.0 $\pm$ 0.2 percent by weight N,N'-diisopropylcarbodiimide conforming to MIL-D-51161 has been added.

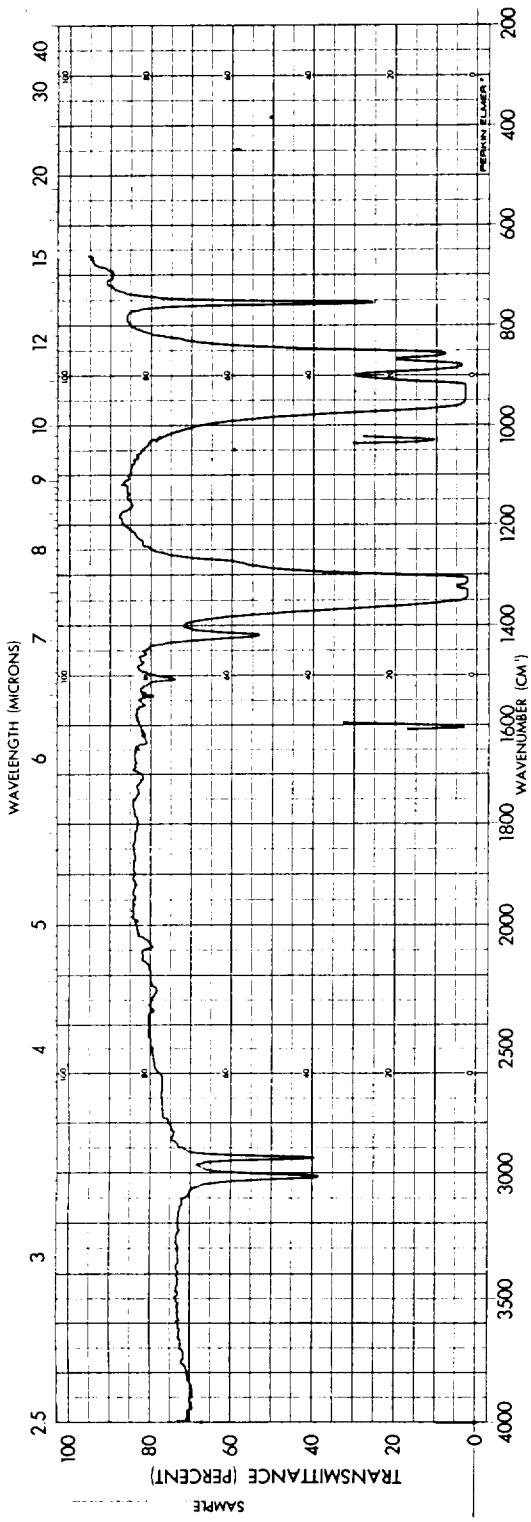
3.3 Preproduction. Prior to the start of regular production of either type of methylphosphonic difluoride, a preproduction lot of type I methylphosphonic difluoride shall be produced in accordance with this specification for test (see 4.3).

3.4 Test report. A test report containing the actual test results for each sample specimen shall be prepared (see 6.2.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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Pathlength: 0.010 mm
Potassium bromide windows
Scan time: 12 minutes
Solvent: neat
Polystyrene calibration peaks:
1601.8 cm^{-1}
1028.3 cm^{-1}

Figure 1. Standard infrared spectrum for unstabilized methylphosphonic difluoride

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4.1.1 Responsibility or compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 Preproduction inspection.

4.3.1 Lotting. A preproduction lot of no less than 225 kilograms of type I methylphosphonic difluoride shall be produced using the same methods, materials, equipment, and processes of manufacture as will be used during regular production.

4.3.2 Sampling (see 6.5). Three representative specimens, in an amount sufficient to accomplish all required tests, but not exceeding 0.5 liter per specimen, shall be randomly taken from the preproduction lot of type I methylphosphonic difluoride and placed in separate clean, dry, nitrogen purged, sealed, high density polyethylene containers labeled to identify the preproduction lot from which they were taken. The three specimens shall then be forwarded for testing to the test facility specified in the contract or order (see 6.2).

4.3.3 Inspection procedure for tests. The preproduction specimens taken as specified in 4.3.2 shall be separately tested as specified in 4.4.4. All three specimens shall be required to pass all tests. If one specimen fails, three additional specimens shall be taken as specified in 4.3.2 and separately tested as specified in 4.4.4; if the three additional specimens pass all tests, the preproduction lot represented shall be accepted.

4.4 Quality conformance inspection.

4.4.1 Lotting. All lots shall be identified and controlled in accordance with MIL-STD-1168.

4.4.1.1 Type I methylphosphonic difluoride. A lot shall consist of the type I methylphosphonic difluoride produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

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4.4.1.2 Type II methylphosphonic difluoride. A lot of type II methylphosphonic difluoride shall consist of a lot of type I methylphosphonic difluoride conforming to 3.1.1 and 3.1.2 which has been stabilized as specified in 3.2.

4.4.2 Sampling.

4.4.2.1 For examination of packaging. Sampling for examination of packaging shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For test (see 6.5). Three representative specimens, in an amount sufficent to accomplish all required tests, but not exceeding 0.5 liter per specimen, shall be randomly taken from the lot of type I methylphosphonic difluoride and placed in separate suitable clean, dry, nitrogen purged, high density polyethylene containers labeled to identify the lot from which they were taken. For type II methylphosphonic difluoride, the lot of type I material to be used to make the type II material shall be sampled.

4.4.3 Inspection procedure.

4.4.3.1 For examination of packaging. The sample unit shall be one filled unit or shipping container, as applicable, ready for shipment. Sample containers shall be examined for the following defects:

AQL 2.5 percent defective

(a) Container not as specified
(b) Container closure not as specified
(c) Exterior of container contaminated with contents
(d) Exterior of container not dry
(e) Seam tape, fiberboard strips, or strapping missing
(f) Foam in place cushioning not as specified (for 5-gallon quantity only)
(g) Marking incorrect, missing, or illegible

AQL 1.0 percent defective

(a) Contents per container not as specified
(b) Container damaged or leaking

4.4.3.2 For test (see 6.2.2). Each sample specimen taken in 4.4.2.2 shall be tested as specified in 4.4.4. All three specimens shall be required to pass all tests. If one specimen fails, three additional specimens shall be taken as specified in 4.4.2.2 and separately tested as specified in 4.4.4; if the three additional specimens pass all tests, the lot represented shall be accepted.

4.4.4 Tests (see 6.5). Water in accordance with ASTM D 1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

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4.4.4.1 Purity by freezing point depression analysis.

(a) Procedure. Use a freezing point apparatus similar to that shown on figure 2. Dry the apparatus along with approximately 1 gram (g) of 50 to 200 mesh coconut charcoal in a vacuum oven at 130° ± 5°C for 6 hours to remove all moisture. Add a few pieces of the dried charcoal to the apparatus and immediately transfer the apparatus to a dry glove box filled with a nitrogen atmosphere and containing a bottle of the specimen. Allow the apparatus to cool to ambient temperature. Add enough specimen to the inner chamber of the apparatus to cover the temperature sensitive portion of the temperature probe by approximately 1 cm. Attach the stirring assembly and seal the shaft of the stirrer so that the inner chamber is sealed from the atmosphere. Remove the apparatus from the glove box and attach the stirring assembly to a variable-speed stirring motor. Insert into the thermowell a device, such as a resistance thermometer or thermocouple, which is capable of measuring temperature to the nearest 0.01°C. Add enough methylene chloride to the well to cover the sensor. Connect the temperature probe to a temperature recorder capable of recording to the nearest 0.01°C. Place the freezing point apparatus in a dry ice-methylene chloride bath and allow the contents to cool to -27°C. At this point apply a vacuum of approximately 2 millimeters (mm) of mercury to the outer chamber while continuing to cool the specimen. Record the cooling of the specimen as temperature versus time. The curve will consist of a steeply sloped line and an after curve produced during crystallization. The freezing point of the specimen is the intersection of the steeply sloped cooling portion of the curve and the extrapolation of the after curve.

(b) Calculation. The freezing point of pure methylphosphonic difluoride is taken as -36.81°C (236.34°K). Calculate the mole percent purity as follows:

$$\log_{10} X = -619.15 \left(\frac{1}{T} - \frac{1}{236.34} \right)$$

where: X = Mole percent purity of methylphosphonic difluoride,
$\log_{10} X$ = Common logarithm of X, and
T = Observed freezing point, °K.

4.4.4.2 Proton NMR response.

(a) Spectrometer specifications.

(1) The spectrometer shall have minimum magnetic field strength of 1.4 Tesla and a suitable internal or external field/frequency lock.

(2) The spectrometer shall have a resolution of 0.5 Hz or less. Resolution is defined as the line width at half-height of a single peak such as tetramethylsilane (TMS) in deuterated chloroform.

(3) The sensitivity of the spectrometer shall be such that the carbon-13 sidebands are readily observed at high amplitude.

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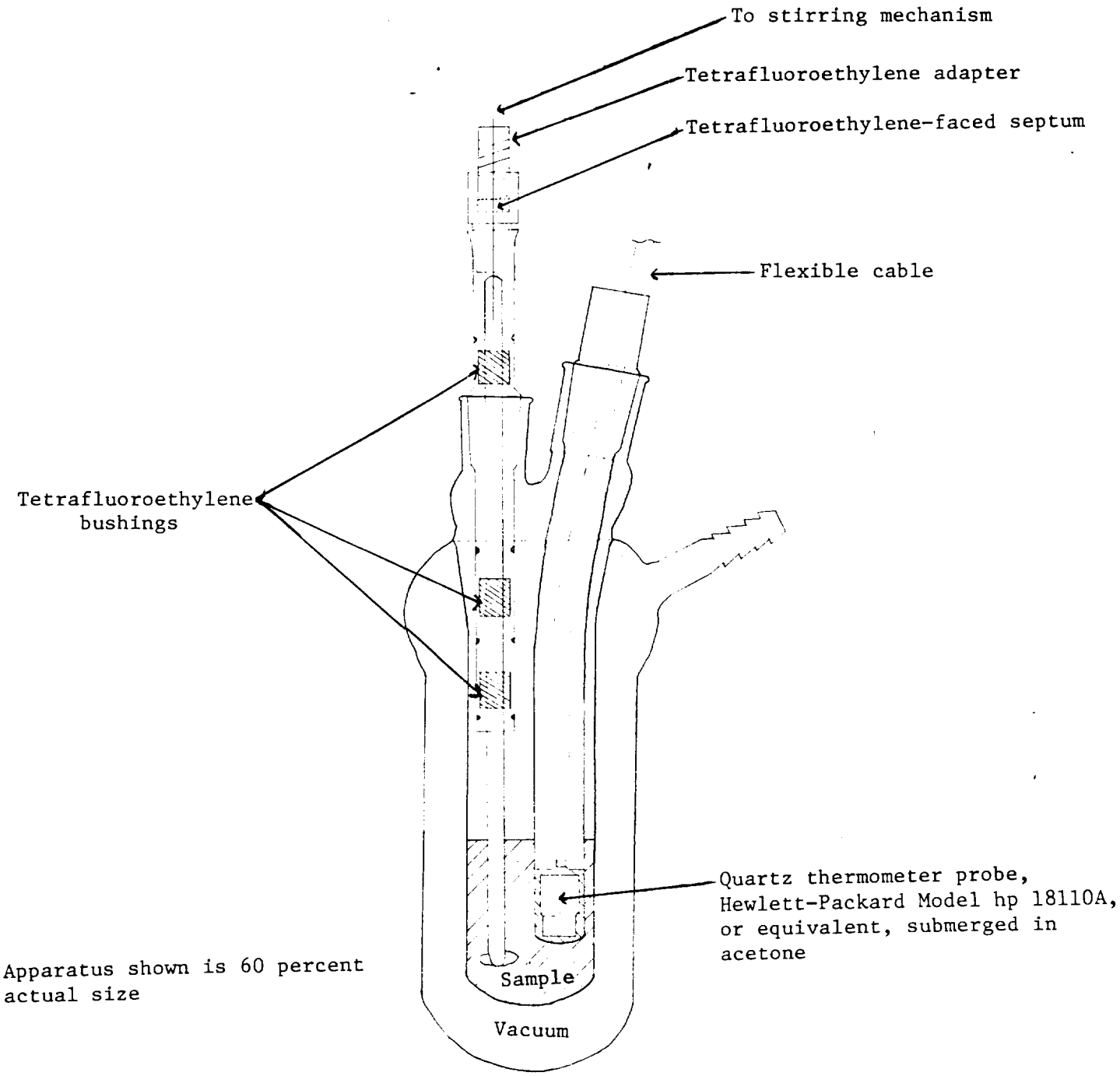


FIGURE 2. Typical freezing point apparatus

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(4) The electronic integral of the spectrometer shall have at least a 1 percent electronic accuracy in the integrating circuit. That is, for the total integral of three groups in the spectrum of a 5 percent solution by volume of ethyl benzene in carbon tetrachloride, the average deviation of the integrals will not exceed 2 percent for five successive scans. For spectrometers that do not have 1 percent electronic accuracy in the integrating circuit but do have 1 percent accuracy in the spectrum amplitude control, (such as a Varian EM-360 NMR Spectrometer), the spectrum amplitude control may be used to increase the integral displacement as described in the procedures outlined below. The integral amplitude control must not be reset after obtaining the initial displacement of the integral in this type of spectrometer.

(5) The spectrometer shall be capable of sweeping from 0 delta, TMS, to at least a downfield value of 17 delta.

(b) Preparation of specimen. Dry the NMR tubes and any glassware to be used in a vacuum oven at 130° ± 5°C for 6 hours to remove all moisture. Store the NMR pressure caps in a desiccator over phosphorus pentoxide prior to use. In a dry glove box filled with a nitrogen atmosphere, transfer 0.4 to 0.5 milliliters (mL) of specimen into a clean, dry 5-mm "Pyrex" or equal NMR tube and seal the tube with a pressure cap and wrap the top of the capped tube with "Parafilm" or equal. Do not add TMS or any other standard reference material to the specimen as contamination of the specimen may result. Make sure that there are no solid particles, including dust, in the NMR tube as this could interfere with the analysis by upsetting the homogeneity.

(c) Procedure. Place the NMR tube containing the specimen in the probe of the spectrometer and spin the specimen. If thermal equilibrium of the specimen to magnet temperature is required, do so first. Use a 500-Hz sweep width or a value as close to 500 Hz as possible. Center the methylphosphonic difluoride resonance at delta 1.94 parts per million (ppm) on the chart paper and tune the instrument for maximum homogeneity. Select an amplitude setting such that the tallest peaks of the methylphosphonic difluoride resonance are 90 to 95 percent of the height of the chart paper. Run the spectrum from delta 17 to delta 0 ppm at a sweep time of approximately 250 seconds. Observe the proton NMR spectrum characteristics. If the resonances appear to be broadened or the normal "ringing pattern" is not observed in the spectrum, check the spectrometer homogeneity using a sample of TMS in chloroform. Proper homogeneity is evidenced by a normal "ringing pattern" and a line width of 0.6 Hz or less (peak width at half-height) for the TMS peak. Improper homogeneity with this sample indicates the need to retune the instrument. If the broadening is not of instrumental origin, it must be assumed that the specimen contains paramagnetic impurities. If the line widths are so broad that the resonance of the methylphosphonic difluoride cannot be recognized as two triplets, then the sample should be analyzed by the infra red method described in 4.4.4.3. Replot the spectrum from delta 17 to delta 0 ppm at the maximum amplitude of the spectrometer keeping the sweep width at approximately 500 Hz and the sweep time at approximately 250 seconds. Determine any spinning sidebands in the spectrum by varying the spinning speed of the specimen tube and replotting the spectrum.

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(d) Determination of impurities. From spectrum plotted in (c) above, determine which peaks are due to carbon-13 sidebands, spinning sidebands and instrumental sidebands. Those peaks which cannot be assigned as sidebands are impurity peaks. (If no peaks other than those attributable to methylphosphonic difluoride and its sidebands are observed, then the sample is pure as far as NMR can determine, and no further work need be done in this section). If the impurity peaks observed in the sample are more than 90 Hz from the main methylphosphonic difluoride resonance and the impurity peaks are large enough so that a reproducible integral may be obtained, use procedure (1) described below. If the impurity peaks observed in the sample are less than 90 Hz from the main methylphosphonic difluoride resonance or if the impurity peaks are so small that a reproducible integral cannot be obtained, use procedure (2) described below.

(1) (See figure 3.) Obtain the electronic integral over the methylphosphonic difluoride resonance, making sure to include all carbon-13 sidebands, spinning sidebands and instrumental sidebands within the integral. (Offset the methylphosphonic difluoride resonance, if necessary). The integral displacement should be 70 to 80 percent of the height of the chart paper. Ingegrate the impurity peaks. If the impurity peaks are small, the integral displacement will be negligible. Increase the integral displacement for the impurity peaks using only the integral amplitude control and/or the spectrum amplitude control. The integral displacement should be increased to at least 10 mm for each impurity peak observed. Take at least three replicate integrals for all of the resonance peaks. These replicate integrals should lie within 2 to 3 mm of each other. Note on the spectrum all integral amplitude values and spectrum amplitude values that were used.

(2) (See figure 4.) Using the electronic integral of the spectrometer, integrate the impurity peaks as well as the lower field carbon-13 sideband of the lower field triplet of the methylphosphonic difluoride resonance. This triplet should be free of all spinning sideband interferences. If reproducible integral values cannot be obtained using the electronic integral, make a copy of the spectrum using a suitable reproduction machine. Carefully cut out the peaks to be integrated from the copy and weigh the peaks on an analytical balance; use the peak weights as integral values.

(e) Calculation of mole percent impurities. If procedure (d)(1) was used, measure the integral displacement obtained for the methlyphosphonic difluoride resonance and divide this value by 3 (methylphosphonic difluoride contains 3 protons). This value is the total methylphosphonic difluoride integral value per proton (T). If the integral displacement for the impurity peaks were increased by adjusting the integral amplitude control and/or the spectrum amplitude control, this displacement must be adjusted to the same conditions used to obtain the integral of the methylphosphonic difluoride resonance. This is done using direct proportions (see figure 3). The adjusted displacement is the correct integral value of the impurity peaks. If procedure (d)(2) was used, divide the integral value (or weight) of the lower field carbon-13 sideband by 3 (methylphosphonic difluoride contains 3 protons). This

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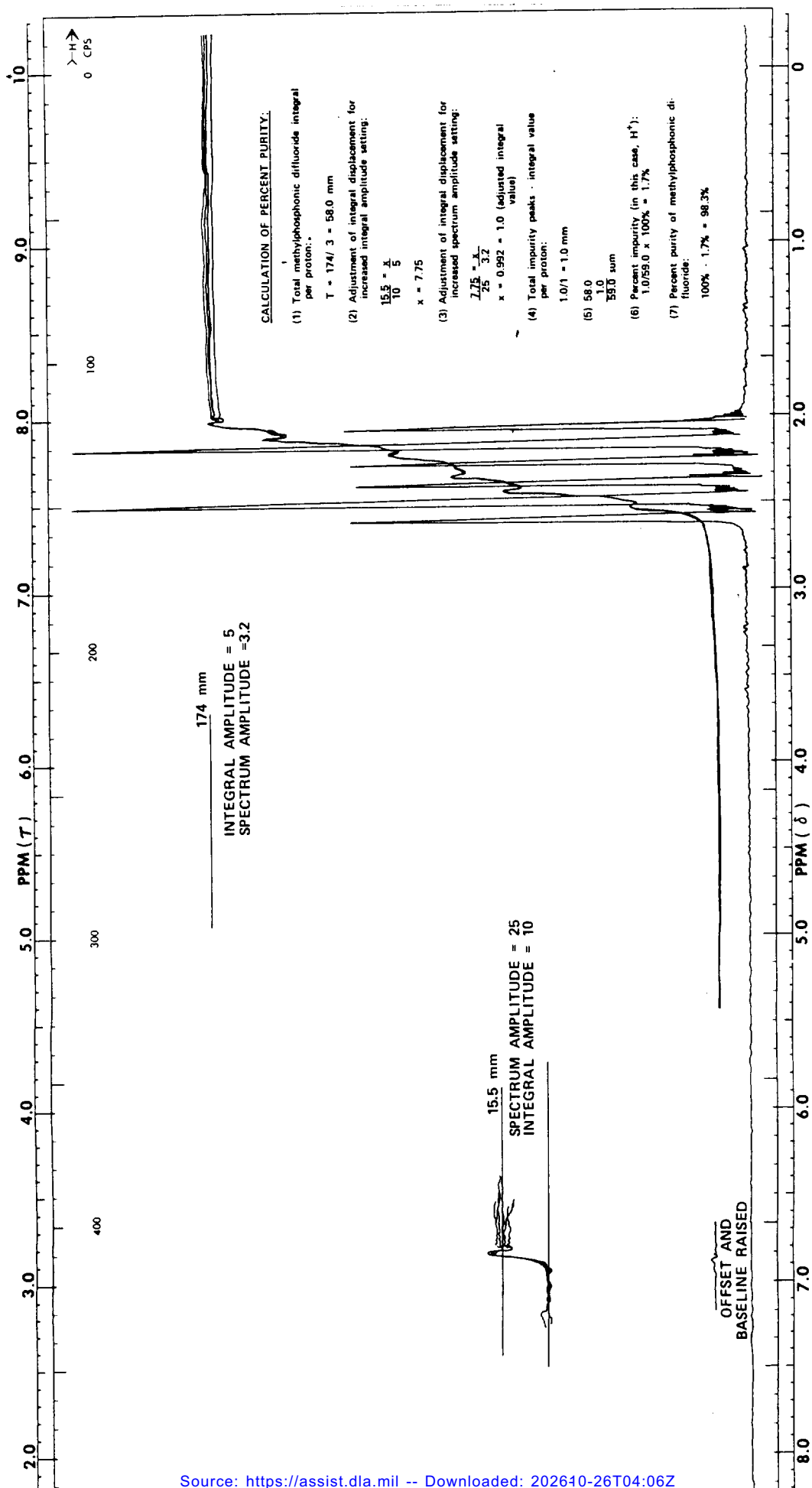


FIGURE 3. Sample NMR proton spectrum for methylphosphonic difluoride

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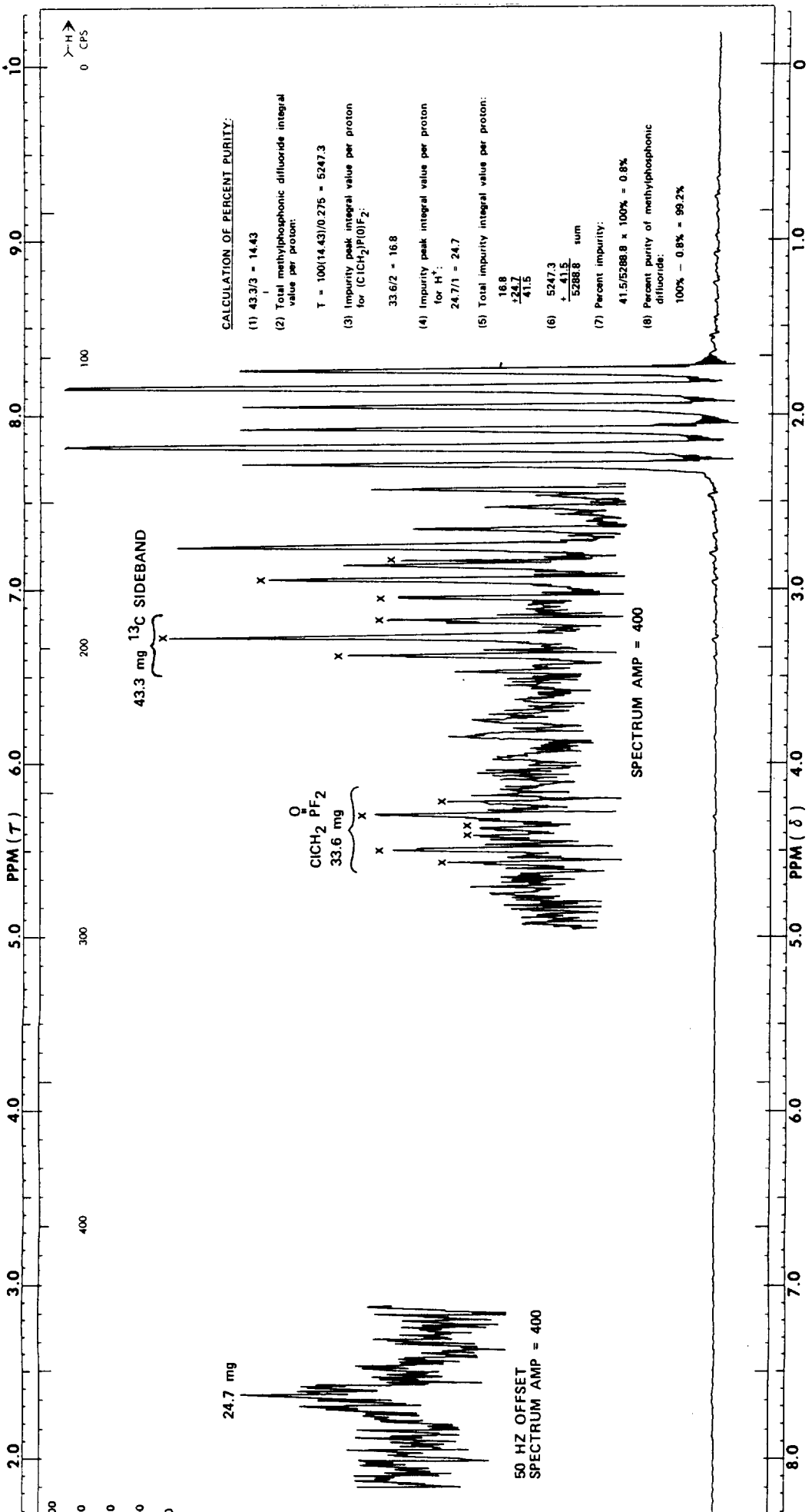


FIGURE 4. Sample NMR proton spectrum for methylphosphonic difluoride

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value represents 0.275 percent of the total integral value per proton of the methylphosphonic difluoride resonance, based on the fact that the natural abundance of carbon-13 is 1.1 percent. Calculate the total methylphosphonic difluoride integral value per proton (T) as follows:

$$T = \frac{100A}{0.275},$$

where: A = The integral value of the lower field carbon-13 sideband divided by 3.

The integral value (or weight) of the impurity is obtained directly from the spectrum (see figure 4). Examine the various impurity peaks present and make a determination as to what compounds they may represent. For each impurity resonance, divide the integral (or weight) of the impurity peak or peaks by the number of protons it represents. If the number of protons represented by the peak or peaks is not known, divide the integral value by 1. (The value obtained after division is the integral value per proton). Add together the value of T calculated above and the integral values per proton determined for each impurity present. Calculate the amount of any impurity present by dividing the integral value per proton for that impurity by the sum obtained in the previous step. Multiply this value by 100 percent to obtain mole percent impurity present (see figures 3 and 4). In order to specifically calculate the amount of hydrogen ion in the specimen, identify the exchangeable proton peak in the spectrum. Determine the integral value (or weight) of this peak and calculate the percent as described above.

NOTE: If there is any question as to whether a particular peak represents an exchangeable proton, complete the entire determination and then add one drop of water to the specimen. Run the spectrum and observe the peak in question. If it grows and moves downfield, it is an exchangeable proton peak.

(f) Calculation of mole percent purity. Calculate the mole percent purity of the specimen by subtracting the total percentage of impurities calculated in (d) from 100 percent (see figures 3 and 4). Report the result to the nearest 0.1 mole percent.

4.4.4.3 Infrared spectrum characteristics. Use a double-beam infrared spectrometer and a sealed cell having a cell path of no more than 0.015 mm or two optical plates with flat, parallel faces mounted in a suitable holder. Scan the spectrum of the empty cell or of the two mounted optical plates between 2.5 and 15 micrometers. No absorption should be present except for periodic "interference fringes" for the sealed cell. If any absorption bands are present, obtain and scan another cell or polish and rescan the optical plates. Working in a glove bag or glove box under dry nitrogen, flush the sealed cell with dry nitrogen and fill it with specimen or prepare a capillary film between the optical plates using a small drop of specimen. In the latter case, a thin uniform film large enough to cover the beam area of the spectrometer is necessary and care should be taken to check for evaporation and atmospheric effects during

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the scan. A dry nitrogen or dry air flush of the spectrometer sample area is recommended. Scan and record the infrared spectrum between 2.5 and 15 micrometers and compare it to the standard spectrum shown on figure 1.

5. PACKAGING

5.1 Unit packing, level B.

5.1.1 One-quart quantity. A quantity of 32 ± 1/2 fluid ounce of type I or type II methylphosphonic difluoride as specified (see 6.2) shall be unit packed in accordance with the applicable requirements of PPP-C-2020 in a nominal 1-quart capacity, type III, class 3, plastic, screwcap container as specified therein. The container shall be formed from polyethylene having a density of 0.941 to 0.965 g per cubic centimeter (cc) and shall conform to MIL-B-26701. Immediately prior to filling, the container shall be purged of air using dry nitrogen conforming to type I, class 2, grade B of BB-N-411. Any spillage of contents on the exterior surface of the closed container shall be washed off with water and then the washed surface shall be dried before further handling. The filled and closed container shall be cushioned with dry, noncombustible, absorbent cushioning material in enclosure B of PPP-C-2020.

5.1.2 One-gallon (gal) quantity. One US gal of type I or type II methylphosphonic difluoride as specified (see 6.2) shall be unit packed as specified in 5.1.1 except that the nominal capacity of the container shall be 1 gal.

5.1.3 Five-gal quantity. Five US gal of type I or type II methylphosphonic difluoride as specified (see 6.2) shall be unit packed in a nominal 5-gal capacity, polyethylene, screw-cap bottle conforming to Department of Transportation (DOT) Specification 34 or 2SL except that the polyethylene of the bottles conforming to Specification 2SL shall have a density of 0.941 to 0.965 g per cc and a melt index of 1.0 g per 10 minutes. Immediately prior to filling, the bottle shall be purged of air using dry nitrogen conforming to type I, class 2, grade B of BB-N-411. Any spillage of contents on the exterior surface of the closed bottle shall be washed off with water and then the washed surface shall be dried before further handling. The shape of the bottle shall be rectangular or cylindrical. The cap of the bottle, after being securely threaded onto the neck, shall be further secured thereto by means of a minimum of two turns of 2-inch wide tape conforming to type III of PPP-T-60.

5.1.4 Fifty-five-gal quantity. A quantity of 55 US gal of type I or type II methylphosphonic difluoride as specified (see 6.2) shall be packed in a nominal 55-gal capacity, polyethylene drum in a steel drum overpack. The polyethylene drum shall conform to DOT Specification 2SL except that the polyethylene shall be a copolymer having a minimum density of 0.950 g per cc and a minimum melt index of 0.30 g per 10 minutes. The steel drum shall conform to DOT Specification 6D with removeable head. The polyethylene drum shall be purged of air as specified in 5.1.1. Any spilled contents shall be removed from the polyethylene drum surface by wiping the surface clean and dry with a cloth. The steel drum shall be securely closed by a bolted locking ring.

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5.2 Packing, level B.

5.2.1 One-quart quantity. One 1-quart container of methylphosphonic difluoride, unit packed as specified in 5.1.1, shall be packed upright in a close-fitting wood box conforming to DOT Specification 15A or 15B. Fiberboard pads formed from material conforming to type CF, class domestic, variety and grade optional of PPP-F-320 shall be inserted between the contents and the inside faces of the box as needed to prevent movement of contents. The filled box shall be closed and strapped as specified in DOT Specification 15A or 15B, as applicable.

5.2.2 One-gal quantity. One 1-gal container of methylphosphonic difluoride, unit packed as specified in 5.1.2, shall be packed upright in a wood box as specified for the 1-quart quantity in 5.2.1.

5.2.3 Five-gal quantity. One 5-gal bottle of methylphosphonic difluoride, unit packed as specified in 5.1.3, shall be packed upright in a wood box as shown on Drawing 400-71-8. When military air shipment is specified (see 6.2), the packed wood box shall be overpacked in a 30-gal capacity, fully removable head, steel drum conforming to DOT Specification 5B or 17C. A 6-inch thick layer of vermiculite conforming to grade 2 or 3 of ASTM C 516 shall be spread in the bottom of the drum. The box shall be placed upright on the vermiculite and additional vermiculite shall be spread around and over the box so that all spaces beside and above the box are filled with vermiculite. The drum shall be securely closed by gasketed cover and bolted locking ring.

5.2.4 Fifty-five-gal quantity. The 55-gal quantity unit packed as specified in 5.1.4 shall require no further protection for shipment.

5.3 Unitization. Uniform quantities of like-sized 1-qt, 1-gal, or 5-gal quantity packs of methylphosphonic difluoride shall be palletized, closure end uppermost, in accordance with the requirements for load type I of MIL-STD-147, using the softwood pallet conforming to type IV of NN-P-71.

5.4 Marking. Unit pack shipping containers and palletized loads shall be marked in accordance with MIL-STD-129 and DOT regulations. The DOT "Corrosive" label and the following precautionary marking shall be applied to each shipping container.

HAZARD:
Corrosive, Eye and Skin
Toxic by Inhalation

DANGER! CAUSES SEVERE BURNS
HARMFUL IF INHALED
REACTS WITH ALCOHOLS TO FORM HIGHLY TOXIC COMPOUNDS

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Do not get in eyes, on skin, on clothing.
Avoid breathing vapor.
Keep container closed.
Use with adequate ventilation.
Wash thoroughly after handling.
Keep away from alcohols.

FIRST AID: In case of contact, flush eyes or skin with plenty of water. Remove contaminated clothing and shoes. If available, treat burns by soaking in iced zephiran, epsom salt or dilute fresh solution of sodium bicarbonate. Magnesium oxide and magnesium sulfate dressings can be used if available. Seek medical attention immediately. If inhaled, remove to fresh air. If not breathing, give artificial respiration. If breathing is difficult, administer oxygen. Seek medical attention.

6. NOTES

6.1 Intended use. Methylphosphonic difluoride is intended for use as a chemical intermediate.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification,
(b) Type of methylphosphonic difluoride required (see 1.2),
(c) Unit quantity required,
(d) If 5-gal quantity should be packed for military air shipment (see 5.2.3),
(e) Testing of methylphosphonic difluoride required for informational purposes (see 6.6), and
(f) Preproduction:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract,
(2) Name and address of test facility and shipping instructions when testing is performed by the Government, and
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR

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Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph No.	Data requirement title	Applicable DID no.	Option
3.4	Test report	DI-T-5247	None

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DOD 5010.12-L, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E 29.

6.5 Sampling and testing precautions. Methylphosphonic difluoride will rapidly react with many common laboratory organic chemicals (particularly alcohols and amines) to produce highly toxic compounds. Extreme caution must be used during sampling, handling, and testing procedures. Personnel involved in these procedures shall follow all applicable safety rules and regulations and should be thoroughly familiar with the chemical reactivity of methylphosphonic difluoride.

6.6 Informational testing. Informational testing requires the testing of the methylphosphonic difluoride specimens for mole percent acidity as specified in 4.4.4.2 and for percent by weight hydrolyzable chloride and parts per million metals as specified in the appendix.

6.7 Subject term (key word) listing.

Difluoromethylphosphine oxide
Methylphosphonic difluoride

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 6810-A024

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APPENDIX

INFORMATIONAL TESTS

10. SCOPE

10.1 Scope. This appendix contains tests to be performed for informational purposes (see 6.2). The results of the tests shall not be used as a basis for accepting or rejecting the lot of methylphosphonic difluoride offered for acceptance. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. TESTS

20.1 Hydrolyzable chloride. Pipet and weigh by difference to the nearest 0.1 mg, a 1 mL sample of methylphosphonic difluoride into a 100 mL volumetric flask. Add 9 mL of chloride-free deionized water and 2 mL of ultrapure (Baker Ultrex or similar purity) nitric acid. Add quantitatively by pipet 10.00 mL of a standard 4000 ppm silver solution prepared from pure silver nitrate. Agitate the flask to assure complete precipitation. Using the same 10 mL pipet, transfer a second aliquot of the 4000 ppm silver solution into a clean 100 mL volumetric flask. Use 10 mL of water and 2 mL of Ultrex nitric acid in place of the sample. Carry both this blank and the sample through the remainder of the procedure. Dilute to mark with chloride-free deionized water and mix well. Transfer a portion of the solution to a centrifuge tube and centrifuge at a minimum of 2400 rpm for 10 minutes. Without disturbing the precipitate, withdraw a 1-mL aliquot of the supernatant liquid and dilute to 100 mL with 5 percent Ultrex nitric acid in deionized water. Prepare a standard curve for silver using the instrument manufacturer's directions. Aspirate the diluted blank and sample to determine their absorbance. Determine the concentration of silver in each solution from the standard curve. Calculate the percent by weight hydrolyzable chloride as follows:

$$\text{Percent hydrolyzable chloride} = \frac{(B - A)\ 0.329}{C}$$

where: B = Concentration in micrograms per milliliter silver in blank,
A = Concentration in micrograms per milliliter silver in sample, and
C = Sample weight in grams.

Should the weight percent chloride be extremely low, the sample size may be increased up to 10 mL without altering the procedure. Should the weight percent be high, a sample should be weighed and diluted and a suitable aliquot used. However, an additional dilution factor must be applied to the formula.

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20.2 Metals.

(a) Preparation of working curves. Prepare separate standard working curves for copper, nickel, iron, and zinc according to the directions in the instrument manual provided with the atomic absorption spectrophotometer used for the analysis. Use glass or plastic ware that has been previously washed in 5 percent nitric acid in water and rinsed with,distilled and deionized water. For each metal ion, plot absorbance versus the respective solution concentration. (Dilute standard solutions at trace levels are not stable and must be remade each day.)

(b) Procedure. Wash three 10-mL flasks in 5 percent nitric acid in water and rinse with distilled and deionized water. Weigh 1 mL of sample to the nearest 0.1 mg, into each of the three 10 mL flasks. Flask 1 should be diluted with deionized water and analyzed for iron. Flask 2 should be diluted with 1 percent ultrapure nitric acid (Baker Ultrex or similar purity acid) in water and analyzed for copper and nickel. Flask 3 should be diluted with 1 percent hydrochloric acid in water and analyzed for zinc. Both sample solutions and standard solutions for each metal should be aspirated into the spectrophotometer consecutively to avoid any baseline shifts in the instrument or variance in flame conditions. A blank should be run using the acid and water and a correction made for any metals in the reagents. Glass flasks and pipets may be used with minimal etching from hydrofluoric acid if they are not allowed to stand but are rinsed immediately after analysis. The concentration of the metal in the sample solution is determined from the standard curve using the following formula:

$$\text{ppm metal} = \frac{\text{micrograms per milliliter metal in solution x 10 mL dilution}}{\text{grams sample}}$$

Concentration of chromium, manganese and molybdenum may be determined in a similar manner. Individual standard working curves must be prepared but the chromium concentration in the sample is determined in deionized water. Therefore, the same flask may be used for iron and chromium analysis. The manganese is determined in 1 percent hydrochloric acid which is the same media used for zinc. Molybdenum presents a problem since it must be diluted with 1 percent ammonium hydroxide. This dilution is more exothermic than the water or acid dilutions and caution must be exercised. Dropwise addition is recommended for safety.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-M-51500A(EA)	880621

3. DOCUMENT TITLE
METHYLPHOSPHONIC DIFLUORIDE, TECHNICAL

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
BARRY L. ELLIOTT	(410) 671-3230	DSN 584-3230
c. ADDRESS (Include Zip Code) USA ERDEC SCBRD ENE S E5027 5232 FLEMING RD APG, MD 21010-5423	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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